

Enhancing Large Language Models with Reinforcement Learning and Human Feedback

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Description

Through this proposal I plan to study the connections between Reinforcement Learning and Large Language Models (LLMs) specifically how Reinforcement Learning techniques can be used to fine tune LLMs for dialog generation. Understanding the connections between RL and LLMs can be highly useful for advancing conversational AI since their integration allows improvement of response quality and aligns models with human intent. While existing LLMs such as GPT and BERT demonstrate outstanding capabilities in understanding and generating content with human intent, fine tuning these models for specific objectives, task performance or ethical standards is a challenging and developing field. These traditional models rely on supervised learning for fine tuning but often this approach does not capture the dynamic and interactive nature of conversations. RL, on the other hand, allows models to optimize responses based on rewards, incorporating user feedback, performance metrics (or task specific goals) and ethical constraints particularly through methods like Reinforcement Learning for Human Feedback (RLHF). The aim of this research proposal is to deep dive into the implementation of such fine tuning. This study is valuable as understanding this connection between the two domains is essential for developing AI systems that are not only intelligent but also ethical, trustworthy, adaptable and user aligned.

Literature Survey

This section highlights some key research and foundational work done in the field of fine tuning LLMs using reinforcement learning. An interesting fact to note is a lot of the work in this field has been contributed by OpenAI and Microsoft. [1] This paper explores the use of Deep Reinforcement Learning to improve dialogue generation by training a sequence-to-sequence model using a policy gradient method. The model is fine-tuned on a reward function that encourages informative, diverse and relevant responses to address the challenges of sparse rewards and sequence generation in dialogue systems, demonstrating improved performance over traditional supervised learning. [2] Introduces reinforcement learning to fine tune a pre-trained dialogue model, DialoGPT. The authors demonstrate using RL from Human Feedback for leveraging reward functions that lead to coherence and context-aware responses. [3] Utilizes RL from Human Feedback to follow instructions more effectively. Models are fine-tuned with human preferences as reward signals. This technique proved to improve the alignment of language models with user intentions. [4] The authors combine supervised learning and RL to train a hybrid dialogue model that optimizes the dialogue policy and goal completion in task-specific dialogue systems. [5] This is another study that uses RL from Human feedback. The researchers focus on training models with ethical guidelines through reward signals that supports harmless and helpful behavior of the AI system while alleviating harmful outputs. [6] This approach uses human preferences to generate a reward model that guides the training process in such a way to produce more desirable and aligned responses for the language model.

References

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